Project Report: Exploring the Changes in Blood Glucose Data and the Impact of Diet on Drug Addicts

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Abstract

This report serves as a detailed documentation of the project, comprehensively summarizing the analysis process and the results of data related to drug addicts. The data sources include Continuous Glucose Monitoring (CGM) blood glucose data, meal records, and psychological assessment forms, among other multidimensional information. This report not only covers each update, but also systematically organizes the overall research progress, relevant experimental results, and preliminary exploratory studies, providing a complete reference for future research and analysis. See the GitHub Repository.

1 Database Introduction

The database was developed as part of a one-year longitudinal follow-up study titled "Personalized Nutrition and Health Program for Individuals in Drug Rehabilitation," a joint initiative by the laboratory at Westlake University and the Zhejiang Drug Rehabilitation Administration. The study focuses on health management and personalized nutrition interventions for individuals undergoing compulsory drug rehabilitation, aiming to explore the long-term effects of scientifically designed nutritional strategies on their physical and psychological well-being.

During the study, 200 voluntary participants were recruited from three compulsory drug rehabilitation centers in Zhejiang Province. Participants underwent regular and systematic assessments, including baseline health data collection, physiological monitoring, nutritional status assessment, psychological health assessment, and lifestyle surveys. This comprehensive dataset integrates physiological, psychological, and behavioral dimensions, providing a valuable resource for precision health management in drug rehabilitation populations. In addition, the data serve as a scientific basis for future public health policies and clinical intervention strategies.

This study was registered at ClinicalTrials.gov: NCT04105621[1] and NCT04060056[2].

1.1 Data Collection Process

Participants were required to meet the following inclusion criteria: **age between 18 and 65 years**, admission to the rehabilitation center **within the past 3 months**, and **no** severe organ failure, infectious diseases (such as hepatitis A, hepatitis B, or Human Immunodeficiency Virus (HIV)), severe mental disorders, or cancer. The data collection process was conducted in three phases: Day 1 (D-1), Day 2 (D-2), and Day 14 (D-14).

On **D-1**, participants first filled out registration forms (including name, ID, identification number, sex, etc.) and received numbered tags along with labeled saliva collection kits. They then proceeded to a room to complete the questionnaire survey. The staffs performed anthropometric measurements, including height, weight, waist circumference, hip circumference, neck circumference, upper arm circumference, calf circumference, body composition, and blood pressure (systolic and diastolic). Then each participant was fitted with a CGM device, which was activated for two minutes. Additionally, participants were provided with a sample collection kit, which included a fecal sample cup, an additional 5ml EP tube, and a urine collection device, as well as a dietary diary. The research team also pre-printed dietary records and standardized cafeteria meal logs based on the scheduled meal plan of the rehabilitation center. Finally, participants were instructed to collect fasting urine and stool samples the next morning.

On **D-2**, the staffs collected fasting urine and stool samples, as well as blood samples.

On **D-14**, staffs assisted participants in removing the CGM device and storing it in designated collection bags, while also retrieving dietary diaries and standardized cafeteria meal records.

The cafeteria staff recorded standardized meal weights over a 14-day period. The staff pre-printed dietary record sheets based on the meal plan for the following two weeks and distributed them to the cafeteria staff. Before cooking, the staff weighed and recorded the total raw weight of each ingredient used (e.g., rice, vegetables, oil, salt). After cooking, they recorded the total cooked weight and then weighed a standard portion of the cooked meal, documenting the data in the log sheet.

The original study design included two follow-up visits, but the current database contains only baseline data from the first visit.

1.2 Key Features

The raw data is stored in the repository: `https://github.com/yichun77/Drug/tree/main/DA_data`, with the details of each table provided in Table 1.

Table 1: Data Content

Table Name	**Main Content**
DA_codebook_full	This table contains all the codes and their corresponding values.
DA_daily_diet	Data on the total amount of food consumed by drug rehabilitation participants daily.
DA_daily_food_clean	Data on the types of food and their nutritional value consumed by drug rehabilitation participants during each meal.
DA_daily_meal	Data on the total nutritional value consumed by drug rehabilitation participants during each meal.
DA_mean_diet	The average nutritional intake of drug rehabilitation participants per day during the study period.
DA_phenotype	Psychological assessment and physiological examination data for both drug rehabilitation participants and normal individuals.
CGM_id_only	Records the blood glucose variations over 14 days for drug rehabilitation subjects wearing CGM devices.

For detailed information on each table, please refer to Fig.1 and the Jupyter Notebooks (`https://github.com/yichun77/Drug/tree/main/codes/`

DA_ID	20	65	69	91	104	106	129	130	148	155	163	171	183	184	193	196
Days				5	7	7			10		0	11	13	14		3
CGM	×	×	×				×	×	×	×	×				×	×
Removed									✓		✓					✓

Figure 1: DA tables information

EDA). In the CGM data, information is missing for 10 individuals undergoing drug rehabilitation, specifically those with IDs 20, 65, 69, 129, 130, 148, 155, 163, 193 and 196. Some participants' meal information records are incomplete, so their data was excluded.

The table named "DA_phenotype" is unique in that it is divided into two sub-tables, representing data for both the healthy population and the Drug Addicts (DA) population. This table includes personal information, psychological assessment scores, and physiological test data. While the data for the DA population is fully complete, the data for the healthy population contains some missing values.

2 Major issues

2.1 Nutritional data check

The daily intake of **22** nutritional components for each individual has been confirmed, and the corresponding figures (e.g. Fig.2 and Fig.3) have been saved in the `results/figures/feature_meal` folder. After inspection, the preprocessed data is complete, with no missing values or outliers. This provides a solid foundation for subsequent research, including nutritional value analysis and meal timing inference.

2.2 CGM data ↔ meal data

The CGM curves for each individual are stored in the folder `results/figures/CGM_curve`. A significant limitation of the dataset is the absence of meal diaries, making it impossible to determine the exact timing of breakfast, lunch,

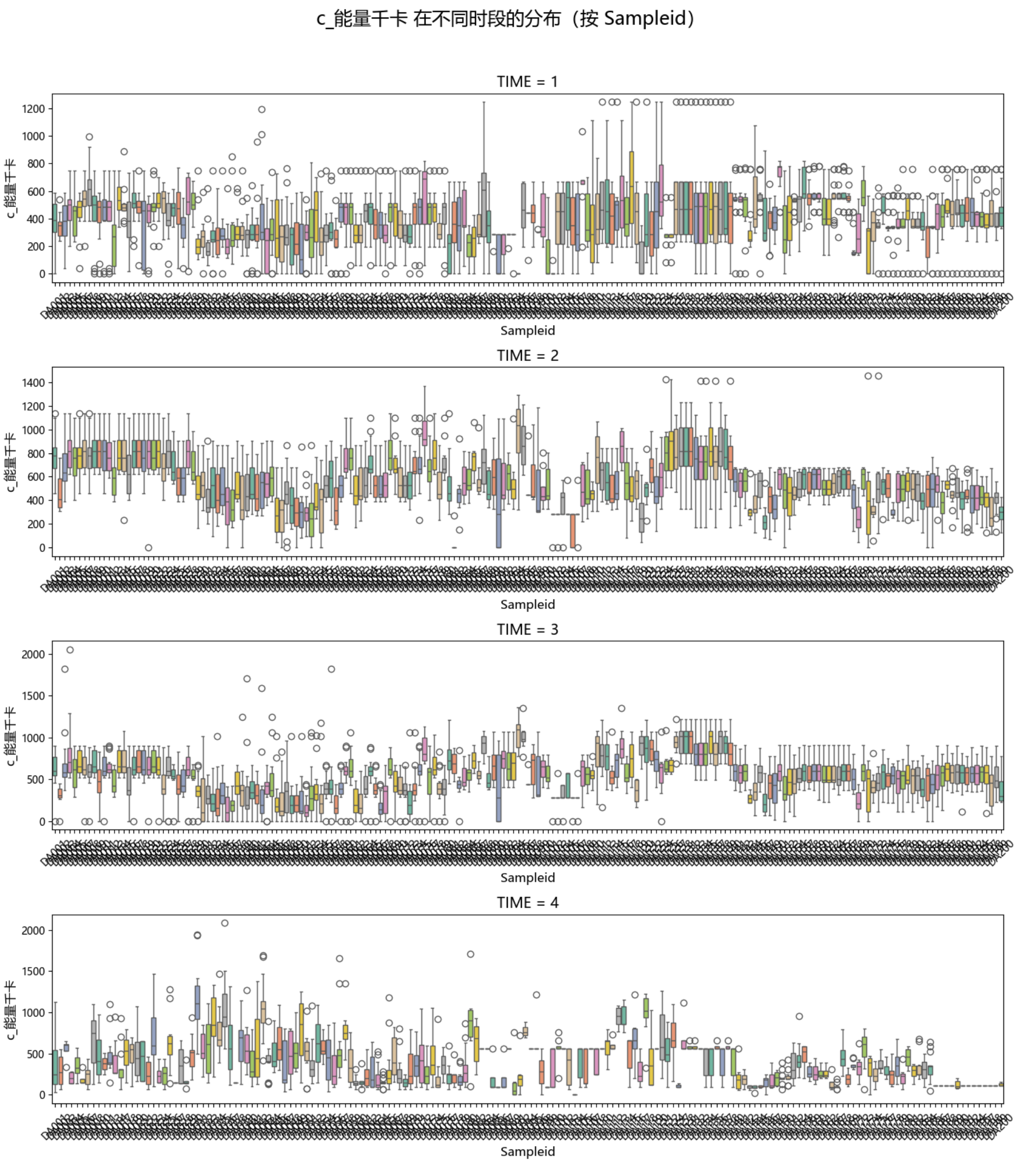

Figure 2: DA Example of nutritional composition 1

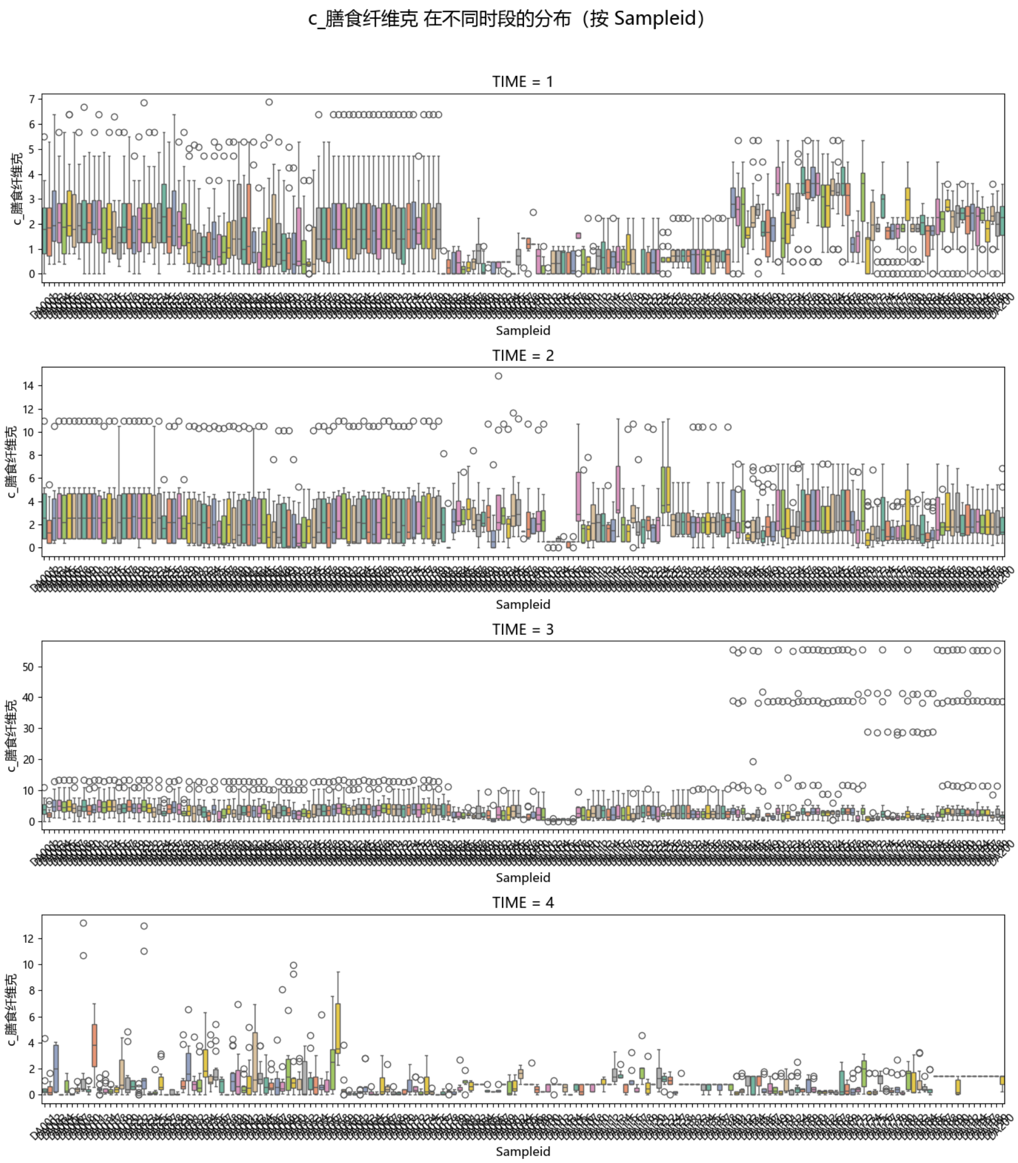

Figure 3: DA Example of nutritional composition 2

dinner, and late-night snacks. As a result, meal times can only be inferred from the CGM data. In this work, we identify potential meal events based on characteristic "glucose spikes"—a postprandial blood glucose response that typically begins 15–30 minutes after food intake, followed by a gradual decline within 1–2 hours.

The main approach involves:

1. Smoothing the CGM curve to reduce noise.
2. Calculating the rate of change in glucose levels (i.e., using differences).
3. Locating the steepest rising points (local maxima of the derivative).
4. These peak points are then clustered into four groups, corresponding to four meals.
5. The representative time of each cluster is output as an estimate of meal timing.

3 xxx

xxxxx

4 xxx

4.1 xxx

xxx

4.2 xxx

xxx

References

[1] ClinicalTrials.gov, Westlake personalized nutrition and health cohort for drug addicts (wepn-da), accessed: 2025-03-25 (2021).
URL `https://clinicaltrials.gov/study/NCT04105621?term=NCT04105621&rank=1`

[2] ClinicalTrials.gov, Westlake precision birth cohort (webirth), accessed: 2025-03-25 (2023).

URL https://clinicaltrials.gov/study/NCT04060056?term=NCT04060056&rank=1